
Darkness Visible: Reading the Exception Handler of a Language Model

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Abstract

1 The final MLP of GPT-2 Small exhibits a fully legible routing program—27
2 named neurons organized into a three-tier exception handler—while the knowl-
3 edge it routes remains entangled across $\sim 3,040$ residual neurons ($3,072 - 27$
4 named = 3,045; we round for readability). We decompose all 3,072 neurons
5 (to numerical precision) into: 5 fused Core neurons that reset vocabulary toward
6 function words, 10 Differentiators that suppress wrong candidates, 5 Specialists
7 that detect structural boundaries, and 7 Consensus neurons that each monitor a
8 distinct linguistic dimension. The consensus-exception crossover—where MLP
9 intervention shifts from helpful to harmful—is statistically sharp (bootstrap 95%
10 CIs exclude zero at all consensus levels; crossover between 4/7 and 5/7). Three
11 experiments show that “knowledge neurons” [Dai et al., 2022], at L11 of this
12 model, function as routing infrastructure rather than fact storage: the MLP ampli-
13 fies or suppresses signals already present in the residual stream from attention,
14 with success scaling with how constrained the context is. A garden-path experi-
15 ment reveals that GPT-2 uses verb subcategorization immediately, with a sugges-
16 tive but non-significant reversal of the classical garden-path effect, consistent with
17 the exception handler operating at token-level predictability rather than syntactic
18 structure. This architecture crystallizes only at the terminal layer—in deeper mod-
19 els, we predict equivalent structure at the final layer, not at layer 11. Code and
20 data: <https://github.com/pbalogh/transparent-gpt2>.

21 1 Introduction

22 The MLP layers of transformer language models are typically treated as opaque nonlinear transfor-
23 mations. Recent work has shown that individual neurons can be interpreted [Gurnee et al., 2023,
24 Bricken et al., 2023], that MLP layers implement key-value memories [Geva et al., 2021], and that
25 specific factual associations can be located and edited [Meng et al., 2022]. But a complete, legi-
26 ble account of an MLP layer’s *routing logic*—readable as pseudocode with named variables—has
27 remained elusive.

28 We provide such an account for Layer 11 of GPT-2 Small, the model’s final MLP, while showing
29 that the knowledge it routes remains distributed across $\sim 3,040$ residual neurons. Our decomposition
30 preserves the original weights to numerical precision while revealing a routing architecture that
31 overturns several assumptions:

- 32 1. **MLP layers are not opaque.** A single exception neuron reliably signals which processing
33 path is active, seven consensus neurons each monitor a distinct linguistic dimension, and
34 20 exception-handler neurons organize into three functional tiers (Figure 1). This routing
35 program is diagnostic, not causal [Balogh, 2026]: the pseudocode captures the functional
36 organization even though no single neuron is a causal bottleneck.

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ROUTING PROGRAM: L11 MLP (27 named neurons)
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consensus = count(N2, N2361, N2460, N2928,
                  N1831, N1245, N2600)
exception = N2123.fires    // 11.3% of tokens

if exception:
    // N2123 is a diagnostic readout, not a causal gate
    CORE(N2123, N2910, N740, N1611, N2044):
        reset vocab -> {the, in, and, a, ,}
        // 54% of output norm, +0.2% PPL
    DIFF(N2462, N2173, N1602, N1800, N2379,
         N1715, N611, N3066, N584, N2378):
        suppress wrong candidates
        repair subword fragments
        // 23% of output norm, +1.3% PPL
    SPEC(N2921, N2709, N971, N2679, N737):
        detect paragraph/clause boundaries
        // 4% of output norm, -0.3% PPL
else:
    // consensus >= 5/7: MLP intervention
    // is counterproductive (dP < 0)
    // but architecture has no bypass

RESIDUAL(3,072 - 27 = 3,045 neurons):
    amplify/suppress attention-derived signal
    // distributed, not individually legible

```

Figure 1: The exception handler as pseudocode. The pseudocode represents the *functional organization* observed in activation patterns, not a literal causal program: N2123 reliably indicates which path is active but does not control it [Balogh, 2026].

2. **“Knowledge neurons” are routing infrastructure.** Neurons identified by Dai et al. [2022] as storing factual knowledge appear across nearly all facts tested—they are high-way signs, not warehouses. At L11 of GPT-2 Small, this reframes ROME [Meng et al., 2022]: editing MLP weights changes routing decisions, not stored facts.
3. **The exception handler has sharp scope boundaries.** The consensus-crossover (§5) confirms that the handler operates at token-level predictability, not syntactic structure; a garden-path experiment provides suggestive but non-significant corroborating evidence (§8, Appendix E).
4. **Routing legibility is unique to the terminal layer.** A survey across all 12 layers shows no comparable structure at any earlier layer—a phenomenon we call *terminal crystallization*, with a testable depth-tracking prediction.

“No light, but rather darkness visible.” —Milton, *Paradise Lost*

Language models are routinely called “black boxes”—as if the darkness inside were uniform. It is not. Inside L11’s MLP we find *structured* darkness: a legible routing program wrapped around knowledge that remains opaque. The darkness is visible precisely because it has structure.

2 Related Work

MLP interpretability. Geva et al. [2021] showed MLPs act as key-value memories; Geva et al. [2023] traced factual recall through attention and MLP layers, finding that mid-layer MLPs promote correct attributes—a more nuanced picture than our binary routing/retrieval framing (§6). Dai et al. [2022] identified “knowledge neurons” in BERT via integrated gradients; we test their claims on GPT-2 and find routing infrastructure, not fact storage (§6.2). Bricken et al. [2023] decomposed MLP activations via sparse autoencoders, extracting monosemantic features from superposed representations. Templeton et al. [2024] extended this to production-scale models (Claude 3 Sonnet), demonstrating that SAE decomposition scales and recovers interpretable features even in large

networks. Our 27 named neurons are legible *without* SAE decomposition—they are individually interpretable because they serve routing functions. The $\sim 3,040$ residual neurons we classify as entangled are precisely the population where SAE methods should prove most valuable: they encode distributed knowledge that resists per-neuron interpretation but may yield to dictionary-learned decomposition. We view our routing/knowledge partition as complementary to the SAE program: we identify *which* neurons are entangled and *why* (they encode content, not control), while SAEs provide the tools to further decompose them.

Circuits and routing. Wang et al. [2023] provided a complete circuit for indirect object identification in GPT-2—tracing a *behavior* across layers; we trace an *entire layer’s* behaviors within L11’s MLP. Balogh [2026] identified the consensus/exception architecture: 7 consensus neurons whose agreement predicts a 94.3pp drop in exception firing, with the MLP becoming counterproductive at full consensus. The present paper extends that from *detecting* routing to *reading* it—characterizing every neuron. Elhage et al. [2021] introduced the residual stream framework; Olsson et al. [2022] identified induction heads; Dettmers et al. [2022] found sparse outlier features compatible with binary routing.

Developmental analysis. Tenney et al. [2019] showed BERT rediscovers the NLP pipeline across layers; Belrose et al. [2023] introduced the tuned lens for layer-wise probing. Our developmental arc extends these to GPT-2’s MLPs with the additional finding that routing legibility concentrates at the terminal layer.

3 Methods

Transparent forward pass. We construct a TRANSPARENTGPT2 wrapping GPT-2 Small (124M parameters) with no weight modifications. At Layer 11, the MLP computes $\text{MLP}(x) = W_{\text{proj}} \cdot h + b_{\text{proj}}$ where $h = \text{GELU}(W_{\text{fc}} \cdot x + b_{\text{fc}})$ is the 3,072-dimensional intermediate activation. We decompose this into tiers by masking h : $\text{Tier}_{\mathcal{S}}(x) = W_{\text{proj}}[:, \mathcal{S}] \cdot (h \odot m_{\mathcal{S}})$, where $\mathcal{S}$ is the neuron index set for each tier and $m_{\mathcal{S}}$ is the corresponding binary mask. The output bias b_{proj} is assigned to the Residual tier (it is a constant offset independent of which neurons fire). Thus $\text{MLP}(x) = \text{Core}(x) + \text{Diff}(x) + \text{Spec}(x) + \text{Residual}(x)$, and the sum equals the original MLP output in the 768-dimensional output space to numerical precision (max elementwise difference $< 6 \times 10^{-5}$, cosine similarity 0.99999994).

Data. All experiments use 512,000 tokens from WikiText-103, processed in the first 500 contiguous sequences of 1,024 tokens from the validation set. Binary firing: $|\text{GELU}(x_n)| > 0.1$; robustness verified across $\theta \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0\}$ (Appendix F).

N2123 activation regimes. At the generic threshold ($|\text{GELU}(x)| > 0.1$), N2123 activates on 70.9% of tokens. However, N2123’s activation distribution is bimodal: a large low-magnitude mode centered near 0.3 and a smaller high-magnitude mode above 1.5. We define the exception-path regime as $\text{GELU}(x_{\text{N2123}}) > 1.0$, which captures 11.3% of tokens and corresponds to the upper mode of this bimodal distribution. This threshold was selected at the minimum density between the two modes; results are robust to variation in the range 0.7–1.5 (Appendix F). Throughout, “exception path active” refers to this high-magnitude regime, while the generic threshold (> 0.1) is used only for binary firing patterns in the enrichment analysis (Table 10).

Statistical methods. Enrichment uses one-sided Fisher’s exact test with Bonferroni correction ($\alpha = 0.05/3072$). All confidence intervals use sequence-level bootstrap (10,000 resamples of 500 sequences, random seed 42) to account for within-sequence autocorrelation. All statistical tests use `scipy.stats` (v1.12).

4 The Exception Handler

Among the 11.3% of tokens where N2123 fires at high magnitude, the remaining neurons organize into three tiers by conditional fire rate (Table 1).

Table 1: Three-tier exception handler. Fire rates conditional on N2123 firing.

Tier	Neurons	Count	Fire Rate	Jaccard	Function
Core	N2123, N2910, N740, N1611, N2044	5	90–100%	≥ 0.91	Vocabulary reset
Diff	N2462, N2173, N1602, N1800, N2379, N1715, N611, N3066, N584, N2378	10	35–88%	0.15–0.89	Candidate suppression, subword repair
Spec	N2921, N2709, N971, N2679, N737	5	14–37%	< 0.15	Boundary detection

Core: a fused mega-neuron. The 5 Core neurons exhibit pairwise Jaccard similarities ≥ 0.91 (peak: 0.998). For context, two *independent* neurons each firing at 95% would produce Jaccard ≈ 0.90 by base-rate overlap alone; the observed 0.998 far exceeds this independence baseline (and the random-init control in §F yields only 0.53). Their output directions all push toward function words: the, in, and, a, ,—a **vocabulary reset** establishing a generic prior before the residual contributes contextual adjustments. The Core accounts for 54% of exception-path output norm but only +0.2% PPL when ablated (Table 3)—a DC offset that the residual stream overwrites. This is a Simpson’s paradox: +2.4% PPL at low consensus (where it matters) and −0.4% at high consensus (where it doesn’t), averaging to the modest +0.2%.

Differentiators: suppression, not promotion. The 10 Differentiator neurons—including a suppression pair (N584+N2378, Jaccard 0.889) and subword repair neurons—contribute 23% of output norm and +1.3% PPL. They suppress wrong candidates rather than promote correct ones, doing the real discriminative work.

Specialists. N737 is a solo paragraph boundary detector (Jaccard < 0.15 with all others). The tier contributes 4% of output norm and slightly *improves* PPL when ablated (−0.3%).

The exception indicator. N2123 detects tokens where the model “doesn’t yet know what’s happening”—it responds to subword fragments and is inhibited by complete content words. It is not causal: zeroing it changes PPL by $< 0.1\%$ [Balogh, 2026]. It is a **vote counter**—a readable summary of the distributed routing decision. Removing the counter does not change the election.

5 Consensus and the Crossover

5.1 Seven Monitors of Normal

The 7 consensus neurons were identified in Balogh [2026] as neurons that (a) fire on $> 75\%$ of tokens overall, (b) show $> 10\times$ enrichment ratio between their most-enriched and most-depleted token classes (Fisher’s exact, Bonferroni-corrected), and (c) have output directions with cosine similarity > 0.4 to the mean MLP output direction at high-consensus positions. We inherit the same 7 neurons here; independent re-identification using the same criteria on our 512K-token dataset recovers all 7. Each monitors a distinct linguistic property (Table 5 in Appendix A). Six content neurons (cos 0.52–0.73) form a **linguistic structure axis**; N2600 (cos ≈ 0.0 with all others) forms a **referential concreteness axis**, firing on currencies, dates, and names while depleting abstract adjectives. Full consensus requires both structural predictability *and* referential concreteness.

5.2 Consensus Predicts MLP Helpfulness

The consensus gradient maps directly to whether L11’s MLP helps or hurts (Table 2).

The crossover between 4/7 and 5/7 is statistically sharp (Figure 2): at 4/7 the MLP helps ($\Delta P = +0.014$, CI $[+0.011, +0.016]$); at 5/7 it harms ($\Delta P = -0.004$, CI $[-0.006, -0.003]$). The routing is a **functional necessity** that correctly identifies when intervention helps versus harms.

Table 2: L11 MLP effect on next-token probability by consensus level (204,800 tokens, sequence-level bootstrap; §3). All CIs exclude zero; crossover at 4–5/7.

Level	Tokens	Mean ΔP	95% CI	
0/7	206	+0.187	[+0.145, +0.231]	▲
1/7	913	+0.094	[+0.079, +0.109]	▲
2/7	3,051	+0.045	[+0.039, +0.051]	▲
3/7	6,283	+0.030	[+0.026, +0.033]	▲
4/7	12,617	+0.014	[+0.011, +0.016]	▲
5/7	34,155	−0.004	[−0.006, −0.003]	▼
6/7	69,625	−0.014	[−0.015, −0.014]	▼
7/7	77,950	−0.020	[−0.021, −0.019]	▼

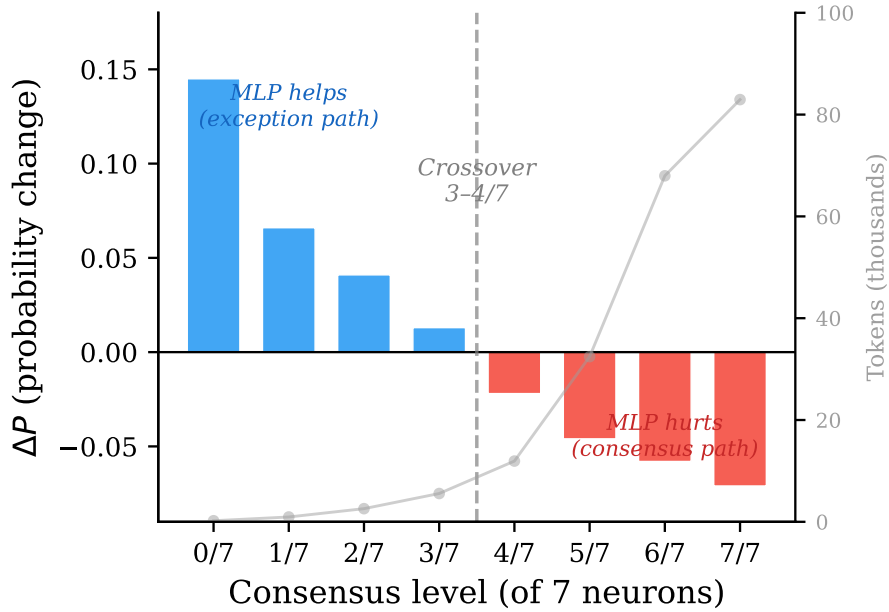


Figure 2: L11 MLP effect by consensus level. Blue: MLP helps. Red: MLP hurts. Crossover between 4/7 and 5/7 matches the causal analysis of Balogh [2026].

143 **Moloch’s compulsion.** ¹ At 7/7 consensus, L11 MLP promotes 7,674 tokens to top-1 while losing
144 3,530—a net gain in accuracy but a net *loss* in average probability (−0.020). The architecture
145 provides no bypass: every token passes through GELU, every distribution gets reshaped. The MLP is
146 architecturally incapable of abstention—a design constraint with implications for efficient inference,
147 since tokens that attention already predicts correctly still pay the full computational cost of MLP
148 processing with no benefit.

149 5.3 Tier-by-Tier Ablation

150 The Differentiators are 6× more important than the Core (Table 3), despite contributing less output
151 norm. Specialists slightly *improve* PPL when ablated—they fire at unpredictable structural bound-
152 aries where their vocabulary push conflicts with the actual next token.

¹In *Paradise Lost*, Moloch counsels perpetual war regardless of outcome—“My sentence is for open war”—preferring action to deliberation. The MLP shares this compulsion: it intervenes on every token, even when intervention is counterproductive.

Table 3: PPL impact of zeroing exception handler tiers (204,800 tokens, sequence-level bootstrap CIs).

Tier	Neurons	PPL	Δ	95% CI
Baseline	—	31.79	—	—
Core	5	31.86	+0.2%	[+0.1%, +0.4%]
Differentiators	10	32.20	+1.3%	[+0.9%, +1.7%]
Specialists	5	31.69	−0.3%	[−0.6%, −0.1%]
All exception	20	32.44	+2.1%	[+1.6%, +2.5%]

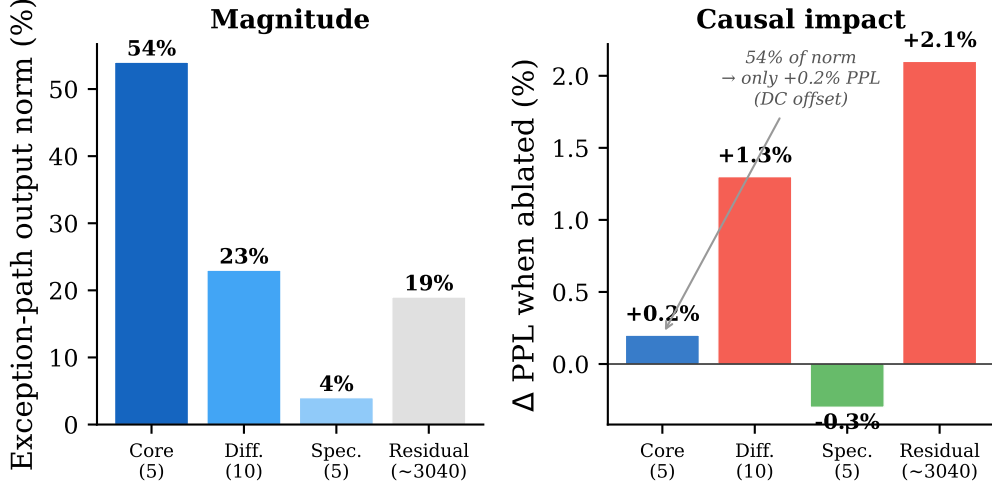


Figure 3: The DC offset paradox. **Left:** Exception-path output norm. **Right:** PPL impact. The Core dominates at 54% of norm but contributes only +0.2% PPL—a DC offset the residual stream already provides.

153 6 Where Knowledge Lives

154 6.1 The MLP Does Not Retrieve Facts

155 For 160 factual cloze prompts spanning 15 categories, we progressively accumulate residual neuron
 156 contributions to test whether factual knowledge is stored combinatorially. Neurons are accumu-
 157 lated in decreasing order of activation-weighted output norm ($|h_n| \cdot \|W_{\text{proj}}[:, n]\|_2$), a target-agnostic
 158 criterion that ranks neurons by generic signal magnitude rather than relevance to any particular an-
 159 swer. In the **static** version (raw W_{proj} columns), the correct token never reaches top-10 (0%). In the
 160 **context-dependent** version (scaled by actual activations), only 18/160 (11%) reach top-10 (median
 161 rank 1,905).

162 The category breakdown reveals the mechanism: highly constrained completions succeed (“300,000
 163 km per $\rightarrow$ *second*,” rank 1; “100 degrees $\rightarrow$ *C*,” rank 1) while arbitrary associations fail (“France is
 164 $\rightarrow$ *Paris*,” rank 6,092; “Germany is $\rightarrow$ *German*,” rank 45,848). This is primarily *amplification* rather
 165 than independent *retrieval*—the MLP provides contextual constraint satisfaction that succeeds when
 166 the constraint space is small but fails when it is large. The 11% success rate for highly constrained
 167 completions represents a limited retrieval capacity, but one dependent on contextual narrowing rather
 168 than stored factual associations. Full category results appear in Appendix B.

169 **Reconciling with mid-layer promotion.** Geva et al. [2023] found that mid-layer MLPs promote
 170 correct attributes during factual recall—a finding that might seem to contradict our claim that L11’s
 171 MLP routes rather than retrieves. The resolution lies in the developmental gradient across lay-
 172 ers: mid-layer MLPs operate on partially formed predictions where the correct answer has not
 173 yet reached high rank, so *promotion* (boosting the correct token’s logit) is the dominant contribu-
 174 tion. By the terminal layer, attention has already assembled a strong candidate set in the residual

Table 4: Exception handler structure across layers (102,400 tokens). Only L11’s known architecture shows tight co-firing.

	L0–L3	L4–L6	L7–L9	L10	L11 (auto)	L11 (known)
Exc. fire rate	6–15%	12–15%	25–36%	38.5%	10.9%	11.3%
Max Jaccard	0.06–0.16	0.13–0.18	0.26–0.36	0.384	0.123	0.998
Enrichment	1.06–1.17	1.11–1.21	1.06–1.09	1.05	1.15	2.0
Hi-Jaccard (>0.5)	0	0	0	0	0	4

stream. L11’s MLP therefore faces a different computational problem—not “which token should be promoted?” but “does the current prediction require correction?”—which is the routing function we characterize. This is consistent with terminal crystallization (§7): the routing program we document exists because the terminal layer inherits a nearly complete prediction and need only intervene when that prediction is wrong.

6.2 Knowledge Neurons Are Routing Neurons

The neurons Dai et al. [2022] identified as “knowledge neurons” are Belial²—they appear, through the eloquence of integrated gradients attribution, to store factual knowledge. They do not. They route.

Attribution overlap. Replicating Dai et al.’s method on 20 prompts at L11, on average 7.3 of each prompt’s top-20 attributed neurons are members of our 27-neuron routing circuit—a $36.5\times$ enrichment over chance ($27/3072 \times 20 = 0.18$). The *same* routing neurons appear across nearly all prompts (N2, N611, N1611, N2044, N2173, N2460, N2600, N2910 each in 16–19 of 20 prompts). If these neurons stored prompt-specific facts, they should differ for “Paris,” “Berlin,” “Jupiter,” and “Au.”

Knockout. Zeroing Dai’s top-20 neurons *increases* target probability by 8.0pp on average—the opposite of storage. These neurons push toward function words; removing them lets factual signal from attention pass through. Only attention head ablation produces the expected decrease (−1.4pp). Knowledge arrives through the residual stream via attention.

Reconciliation. Integrated gradients conflates *influence* with *storage*. These neurons have high gradients because they are control points: changing a highway sign has large causal effects without the sign “storing” destinations. Whether Dai et al.’s findings in BERT—a bidirectional model with fundamentally different information flow—represent genuine storage or also reduce to routing remains an open question. Full details in Appendix C.

7 Terminal Crystallization

If the routing program reflects a general organizational principle of MLPs, it should appear across multiple layers. If it is unique to the terminal layer, it reveals something about the model’s developmental trajectory. We replicated the full characterization at all 12 layers (Table 4). The highest Jaccard similarity found at *any* other layer is 0.384 (L10). L11’s Core achieves 0.998—a qualitative gap. No other layer shows Specialist neurons, suppression pairs, or enrichment above $1.21\times$.

Testable prediction: Legible routing crystallizes at L11 because it is the **last opportunity** to adjust predictions. In deeper models (GPT-2 Medium, 24 layers; Large, 36 layers), equivalent structure should appear at the final MLP, not at L11.

The logit lens and tuned lens [Belrose et al., 2023] confirm a three-phase developmental arc—scaffold ($\sim 1\text{--}2\%$ top-1 gain/layer), decision ($\sim 5\text{--}6\%$ /layer at L7–L9), terminal—under both probing methods (Appendix D).

²In *Paradise Lost*, Belial “could make the worse appear / The better reason”—persuasive but misleading.

8 Discussion

The program and the database. Our decomposition separates L11’s MLP into a legible **program** (27 named neurons: binary routing) and an opaque **database** (~3,040 residual neurons: contextual adjustment). The analogy is a software system where the control flow is open-source but the database is encrypted.

Scope of the exception handler. The handler detects token-level vocabulary uncertainty but not syntactic reparse. A garden-path experiment (15 minimal pairs; Appendix E) reveals a suggestive but non-significant trend: GPT-2 uses verb subcategorization immediately (“struggled” cannot take an object), so the intransitive condition produces lower surprisal at disambiguation in 8 of 15 pairs ($W = 52$, $p = 0.65$, median $\Delta = +0.3$ bits). N2123 shows no differential response ($t(14) = -0.45$, $p = 0.66$; powered for $d \geq 0.78$). The exception handler fires on vocabulary uncertainty, not parse ambiguity. This parallels human sentence processing: Trueswell et al. [1993] demonstrated that readers use verb subcategorization information immediately, with garden-path effects occurring only for NP-bias verbs (e.g., *forgot*) but not for S-bias verbs (e.g., *insisted*)—the same asymmetry GPT-2 exhibits.

Efficiency. Since L11’s MLP is counterproductive at high consensus (Table 2: $\Delta P = -0.020$ at 7/7), bypassing it for the 40% of tokens at full consensus would save the MLP forward pass while *improving* prediction quality on those tokens—a routing-aware alternative to speculative decoding. The aggregate PPL cost of such selective bypass is an empirical question we leave to future work.

Model editing. “Knowledge neurons” are routing infrastructure, reframing ROME [Meng et al., 2022]: weight edits change routing decisions, not stored facts. This predicts edits should generalize across phrasings but fail across domains where routing structure differs.

Limitations. Single model (GPT-2 Small), single domain (WikiText-103), limited garden-path stimuli ($N = 15$), and an underpowered transplant experiment ($N = 5$). Full discussion in Appendix G.

9 Conclusion

The final MLP of GPT-2 Small exhibits a legible routing program: 7 consensus neurons detect “normal language” along distinct linguistic dimensions, a single exception neuron indicates which of two processing paths is active, and ~3,040 residual neurons provide contextual adjustments to knowledge already in the residual stream. This architecture crystallizes only at the terminal layer.

The MLP’s contribution is not knowledge retrieval but **routing**: deciding whether to intervene or abstain. We can read 27 neurons as a routing program. The 3,040 neurons they route remain opaque.

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295 A Consensus Neuron Characterization

Table 5: Consensus neuron specializations (512K tokens).

Neuron	Dimension	Rate	Key Evidence
N2	Clausal continuation	88.4%	Fires on and, but, also mid-clause; silent at clause boundaries
N2361	Syntactic elaboration	84.1%	Fires on that, while, neither, fully; depleted on There, United
N2460	Relational embedding	86.0%	Fires on an, into, per, other in prep. phrases; depletes According to 3.3%
N2928	Sequential structure	91.4%	Fires on ordinals, lists, parallel structure; highest mean activation (0.79)
N1831	Discourse coherence	81.0%	Fires on topic-continuing phrases; 77K disagreement tokens
N1245	Argument structure	85.5%	Fires on leadership, validity, one of the; marks semantic roles
N2600	Concrete reference	79.2%	Fires on \$, dates, names; depletes natural (1.8%), social (3.1%), abstract adj.

296 The six content neurons (N2–N1245) have pairwise cosine similarities of 0.52–0.73 in output direc-
 297 tion space: aligned but not redundant. N2600 is nearly orthogonal ($\cos \approx 0.0$), forming a separate
 298 axis. Full consensus therefore requires both structural predictability *and* referential concreteness.

299 The exception neuron N2123 is aligned with the consensus mean (cosine 0.838): both routing paths
 300 push toward the same safe vocabulary—the difference is activation context, not direction.

301 **Consensus as linguistic predictability.** Tokens at 0/7 consensus are 27% paragraph breaks and
 302 rare subwords; at 7/7 they are dominated by punctuation and function words. The relationship holds
 303 after controlling for frequency: pre-MLP top-1 probability monotonically decreases from 0.249 (7/7)
 304 to 0.057 (0/7), tracking contextual confidence, not token frequency.

305 B Knowledge Extraction: Full Results

306 **Progressive prediction.** For “In 1969, astronauts landed on the _____”: **Before MLP:** residual
 307 predicts moon (from attention). **After Core:** shifts toward the (vocabulary reset). **After Differ-**
 308 **entiators:** wrong candidates suppressed. **After Residual:** returns to moon with redistributed mass.
 309 Across 12 prompts, 10/12 show this pattern; the 2 exceptions involve subword-initial predictions.

310 C Knowledge Neurons: Full Details

311 **Attribution overlap details.** The $36.5\times$ enrichment uses a uniform baseline ($27/3072 \times 20 =$
 312 0.18). Since top-20 integrated-gradient neurons are selected for high causal influence, and routing
 313 neurons have high influence by definition, the true enrichment over a causally-matched baseline
 314 would be lower. The qualitative finding—that the *same* routing neurons recur across diverse facts—
 315 is the more robust evidence.

316 **Knockout details.** Of the 20 prompts, 17 showed increased target probability when Dai’s neurons
 317 were zeroed (range: +0.3pp to +24.3pp), 2 showed negligible change ($< \pm 0.1$ pp), and 1 showed
 318 a decrease (−2.1pp). The consistency across diverse facts (85% showing improvement) confirms
 319 that these neurons systematically suppress factual signal rather than occasionally doing so. The
 320 apparent tension between knockout improving factual recall (+8.0pp) while full exception ablation
 321 worsens PPL (+2.1%) resolves because the handler is optimized for the common case (function-
 322 word prediction at low consensus) at the cost of rare factual completions.

Table 6: Knowledge retrieval across 160 factual prompts.

Prompt (abbreviated)	Target	Base	Static	Context
<i>Retrievable (context rank ≤ 10):</i>				
... 300,000 km per	second	1	244	1
... 100 degrees	C	3	68	1
... created by Linus	Tor	1	2,551	1
... London occurred in	16	4	372	2
... begins with four	notes	2	3,159	2
<i>Not retrievable (context rank $> 1,000$):</i>				
... France is	Paris	5	3,429	6,092
... language of Germany	German	4	1,402	45,848
... Russia is	Moscow	7	5,926	33,592
... blue	whale	1	22,915	26,557
... food from	Japan	2	2,731	48,186

Category	n	Top-10	Median rank
Historical events	10	4	16
Physics	10	4	93
Music	10	1	554
Mathematics	10	1	594
Technology	10	3	764
Chemistry	10	1	887
Astronomy	10	1	1,074
Animals	10	1	1,616
Historical people	15	1	1,812
Geography (location)	10	0	2,316
Biology	10	0	3,041
Literature/language	10	0	3,240
Food	10	0	8,100
Capitals	15	0	14,960
Languages (country)	10	1	26,530
All (160)	160	18 (11%)	1,858

Table 7: Integrated gradients overlap with routing neurons (20 prompts).

	In top-20	Expected by chance
Consensus neurons (7)	2.6 avg	0.05
All routing neurons (27)	7.3 avg	0.18
Enrichment	36.5×	—

Activation transplant. For 5 country-capital pairs, transplanting Dai’s top-20 neuron activations from source to destination prompt transplanted no facts (0/5). Source-fact probability changed < 0.02 pp; this experiment ($N = 5$) is illustrative, not standalone evidence.

D Logit Lens and Developmental Arc

The tuned lens corrects early-layer underestimates (mean +6.3pp at L0–L3) while converging at L11 (+0.0pp). The developmental arc is confirmed: decision-phase layers gain 5.3pp/layer under tuned lens vs. 1.5pp in scaffold phase.

Illustrative cases. **Easy:** “Abraham → Lincoln” locks in at L0; L11 MLP slightly reduces confidence. **Medium:** “moon” overtakes “planet” at L10’s MLP. **Hard:** “second” (speed of light) reaches top-1 only at L11.

Table 8: Top-1 accuracy via logit and tuned lens [Belrose et al., 2023] (204,800 tokens). “First lock-in (Tuned)” = percentage of tokens that first reach top-1 accuracy at this layer under the tuned lens.

Layer	Logit Top-1	Tuned Top-1	Δ	First lock-in (Tuned)
Emb	0.7%	3.5%	+2.8	3.5%
L0	3.0%	9.0%	+6.0	6.2%
L1	3.9%	9.7%	+5.8	1.2%
L2	3.9%	10.7%	+6.7	1.4%
L3	5.0%	11.7%	+6.6	1.4%
L4	5.9%	12.6%	+6.7	1.3%
L5	8.6%	16.2%	+7.6	4.0%
L6	10.8%	19.0%	+8.2	3.4%
L7	17.0%	24.6%	+7.6	5.9%
L8	22.7%	29.7%	+6.9	6.0%
L9	29.5%	34.9%	+5.4	5.9%
L10	34.4%	37.3%	+2.9	3.7%
L11	39.0%	39.0%	+0.0	3.0%
Never	—	—	—	53.0%

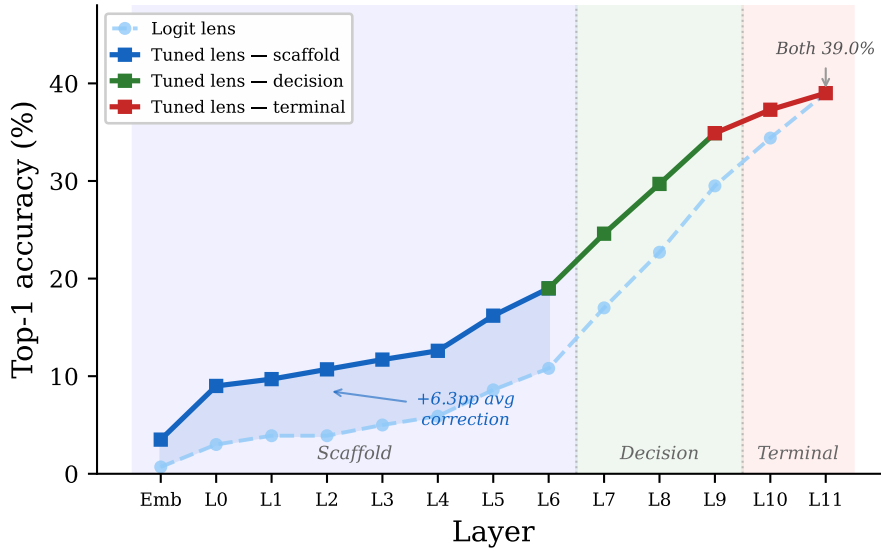


Figure 4: Top-1 accuracy across layers. The developmental arc holds under both logit lens (dashed) and tuned lens (solid).

333 **L11H7: the dominant attention head.** L11’s most important head ($6\times$ the next) sends 45.4% of
334 weight to BOS—the attention sink phenomenon [Xiao et al., 2024]. Exception tokens attend to BOS
335 even more (47.0% vs 37.3%).

336 E Garden-Path Experiment

337 Following Trueswell et al. [1993] and Van Gompel and Pickering [2001], we constructed 15 minimal
338 pairs (Table 9). In each pair, the intransitive verb cannot take a direct object, forcing the post-verbal
339 NP to be parsed as a new clause subject. The transitive verb is ambiguous: the post-verbal NP could
340 be its object.

341 **Example.**

Table 9: All 15 garden-path minimal pairs with surprisal at disambiguation. Pairs 1–5 follow Van Gompel and Pickering [2001]; pairs 6–15 extend the set.

#	Intrans. verb	Trans. verb	Disambig.	S_{int}	S_{trans}
1	struggled	scratched	took	5.4	11.6
2	sneezed	visited	prescribed	6.8	16.2
3	dozed	watched	next	8.9	11.5
4	escaped	attacked	searched	17.0	19.4
5	slept	ignored	standing	13.2	11.3
6	cried	woke	in	9.0	6.9
7	purred	bit	sitting	14.5	14.2
8	fainted	alarmed	on	9.3	7.5
9	galloped	threw	on	5.4	5.7
10	erupted	destroyed	of	1.6	1.3
11	sputtered	startled	under	12.8	11.9
12	lied	accused	at	6.9	8.2
13	performed	impressed	in	5.0	5.3
14	marched	disobeyed	at	7.4	8.5
15	blazed	trapped	near	12.2	8.3
Median Δ (trans – int)				+0.3 bits	

342 *After the dog **struggled** the vet took off the muzzle.*
 343 *After the dog **scratched** the vet took off the muzzle.*

344 In the intransitive version, human readers initially parse “the vet” as the object of “struggled,” then
 345 reparse at “took.” We measured surprisal, consensus, N2123, and MLP delta at disambiguation.

346 **Results.** No differential N2123 response (0.102 vs 0.105, $t(14) = -0.45$, $p = 0.66$; powered for
 347 $d \geq 0.78$). The transitive condition produces higher surprisal at disambiguation in 8 of 15 pairs,
 348 but this trend is not statistically significant (Wilcoxon $W = 52$, $p = 0.65$, median $\Delta = +0.3$
 349 bits). The two strongest pairs (“scratched” +9.4, “struggled” +6.2) drive the positive mean (+0.8
 350 bits), but the effect is inconsistent across the set. GPT-2 uses verb subcategorization immediately:
 351 “struggled” cannot take an object, so “the vet” is parsed as a new-clause subject from the start.
 352 Where subcategorization is unambiguous, the garden-path is *reversed*: transitive verbs create the
 353 reparse. However, the small and variable effect size (8/15 pairs, $p = 0.65$) means this reversal is
 354 hypothesis-generating rather than confirmatory, consistent with our framing in §??.

355 **Connection to constraint-based parsing.** Trueswell et al. [1993] separated lexical preference
 356 effects from true garden-paths, showing that verb subcategorization is used immediately in hu-
 357 man parsing—contra lexical-filtering models where subcategorization is only consulted after initial
 358 structure-building [see also Traxler and Tooley, 2007, for related evidence on lexical mediation].
 359 GPT-2’s N2123 null result is consistent with this picture: verb-specific constraints appear to be ap-
 360 plied at the verb itself rather than deferred to a later reanalysis stage, though the non-significant
 361 surprisal effect ($p = 0.65$) means this remains a hypothesis rather than a confirmed finding. Britt
 362 et al. [1992] showed that discourse context can override shallow attachment preferences but not
 363 deep clause-level reparse. Our verb subcategorization ambiguity is structurally analogous: GPT-2
 364 resolves it immediately without exception-handler involvement, paralleling both Trueswell et al.’s
 365 constraint-based account and Britt et al.’s autonomous syntactic component.

366 **F Controls**

367 **Null model.** A randomly initialized GPT-2 (same architecture, untrained weights) shows no struc-
 368 ture:

Metric	Trained	Random init
Exception fire rate	11.3%	~64%
Core max Jaccard	0.998	0.53
Consensus–exception anticorrelation	94.3pp range	<2pp (flat)
Enrichment (max)	2.0×	1.02×

Threshold robustness. Core co-firing (Jaccard ≥ 0.45) and consensus-exception anticorrelation (spread ≥ 0.11) persist across $\theta \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1.0\}$. No threshold produces qualitatively different structure.

Table 10: Enrichment of routing neurons under exception firing (Fisher’s exact, Bonferroni-corrected).

Neuron	Tier	Base rate	Exc rate	Enrichment	p -value
N2123	Core	70.9%	100.0%	$1.41\times$	$< 10^{-300}$
N584	Differentiator	6.2%	8.3%	$1.33\times$	$< 10^{-300}$
N2378	Differentiator	5.7%	7.4%	$1.30\times$	$< 10^{-300}$
N737	Specialist	4.0%	4.9%	$1.22\times$	$< 10^{-300}$
N1602	Differentiator	22.7%	24.4%	$1.08\times$	$< 10^{-300}$
N611	Differentiator	67.2%	71.8%	$1.07\times$	$< 10^{-300}$
Consensus (5/7)		—		0.96–1.01	> 0.05
Remaining (15)		—		0.98–1.01	> 0.01

G Limitations

- **Single model:** All analysis is on GPT-2 Small (124M). Terminal crystallization in deeper models is predicted but untested.
- **Single domain:** WikiText-103 (encyclopedic prose). Consensus characterizations may not hold for dialogue, code, poetry, or legal text.
- **Single layer depth:** L11 is characterized comprehensively; L7 and L10 only sketched.
- **Garden-path stimuli:** 15 pairs testing verb subcategorization only; reduced relatives may differ.
- **BPE confounds:** Disambiguation words may fall at different absolute positions due to tokenization.
- **Transplant:** $N = 5$; treated as consistency check only.
- **SAE alternative:** We characterize neurons individually rather than via sparse autoencoders [Bricken et al., 2023, Templeton et al., 2024], which might resolve additional structure within the $\sim 3,045$ residual neurons our manual method classifies as entangled.

H Broader Impact

This work contributes to mechanistic interpretability by providing a replicable template for decomposing MLP layers into functionally distinct neuron groups. The exception-handler framing may improve safety monitoring: if harmful content generation is mediated by specific neurons’ routing decisions, the consensus/exception decomposition could help localize and audit those pathways. Our finding that model editing (ROME) modifies routing rather than stored knowledge has implications for alignment: interventions targeting “knowledge editing” may have different behavioral consequences than intended. We note the usual dual-use concern that better mechanistic understanding could also facilitate adversarial manipulation, though we believe the interpretability benefits outweigh this risk at GPT-2 Small’s scale.